"Derivation" of Taylor Series

For any smooth, continuous, and differentiable function $f(x)$, we note that:

$$① \quad f(x) = f(0) + \int_0^x f'(t)\,dt$$

by definition.

Now write the integral to allow us to integrate by parts:

$$② \quad f(x) = f(0) + \int_0^x f'(x-t)\,dt$$

$$\boxed{\begin{aligned} &\int u\,dv = uv - \int v\,du \\ &u = f'(x-t) \quad dv = 1 \\ &du = f''(x-t) \quad v = t \\ &\int_0^x f'(x-t)\,dt = t f'(x-t)\Big|_0^x + \int_0^x t f''(x-t)\,dt \end{aligned}}$$

Now integrate by parts:

$$③ \quad f(x) = f(0) + f'(0)x + \int_0^x t f''(x-t)\,dt$$

... and integrate by parts again:

$$④ \quad f(x) = f(0) + f'(0)x + \frac{1}{2} f''(0)x^2 + \frac{1}{2}\int_0^x t^2 f'''(x-t)\,dt$$

in general:

$$\boxed{f(a) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n}$$